Unified ILP Models for Bicycle Station Placement

COVER: 2nd Consortium Meeting Progress Report

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Prologue

House of Problems (1.1): Problem Description

"Given a set of candidate bicycle station locations, the objective is to select a subset of stations that maximizes accessibility to public transportation, while ensuring that a minimum proportion of demand in each zone is satisfied, a minimum number of parking stations is provided per zone, and the total budget constraint is not exceeded."

Existing Solution

Jenn-Rong Lin and Ta-Hui Yang. Strategic design of public bicycle sharing systems with service level constraints. Transportation Research Part E: Logistics and Transportation Review, 47(2):284–294, 2011.

Related Work

Related Work

- **Maximal Covering Location**
- **Flow Termini Coverage Model**
- **PT Integration**
- **BSS Planning Tools**

The Gap

No existing model combines **flow-termini coverage** with PT integration as a **planner decision**

Model: Parameters

Parameters	**Unit / Domain**	**Description**
$R_{\text{bss}}^{\text{walk}}$	(m)	Max. walking radius to/from BSS station
$R_{\text{pt}}^{\text{walk}}$	(m)	Max. walking radius from BSS station to PT stop
$R_{\text{dir}}^{\text{ride}}$	(m)	Max. cycling radius for direct BSS-to-BSS trip
$R_{\text{pt}}^{\text{ride}}$	(m)	Max. cycling radius to PT-adjacent BSS station
$T_{\text{max}}^{\text{pt}}$	(min)	Max. PT travel time between two stops
α	$[0, 1]$	Proportion of stations to open (budget proxy)
$w_1,\ w_2$	$(w_2 \geq w_1 \geq 1)$	**Control:** tier coverage weights

Model: Coverage Tiers

Given a flow q with origin o_q and destination d_q, let:

Tier 1 — Direct (y_q^1)

$$o_q \xrightarrow{R_{\text{bss}}^{\text{walk}}} j \xrightarrow{R_{\text{dir}}^{\text{ride}}} m \xrightarrow{R_{\text{bss}}^{\text{walk}}} d_q$$

Walk $\rightarrow$ bike $\rightarrow$ walk

Tier 2 — PT-Hop (y_q^2)

$$o_q \xrightarrow{R_{\text{bss}}^{\text{walk}}} j \xrightarrow{R_{\text{pt}}^{\text{ride}}} i \xrightarrow{R_{\text{pt}}^{\text{walk}}} k \xrightarrow{\text{PT}} k' \xrightarrow{R_{\text{bss}}^{\text{walk}}} d_q$$

Bike + PT

Tier 3 — Multimodal (y_q^3)

Tier 1 $\cap$ Tier 2

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AND
Bike + PT

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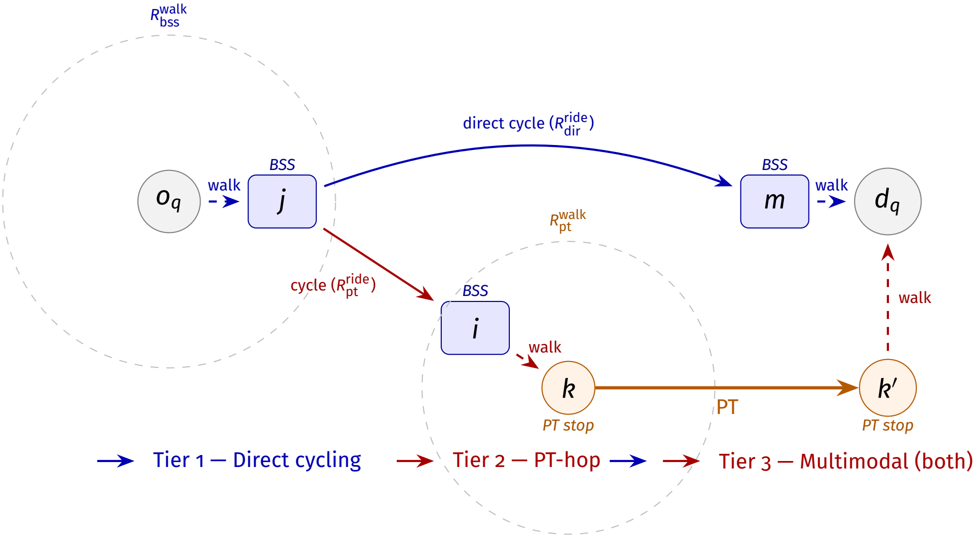
Representative Diagram



Model: Objective & Policy Cases

Objective Function

$$\max \sum_{q \in Q} F_q \left[y_q^1 + (w_1 - 1)\, y_q^2 + (w_2 - w_1)\, y_q^3 \right]$$

Policy Cases

Case	(w_1, w_2)	**Behavior**
1	$w_1 = w_2 = 1$	Direct only — recovers FTCM
2	$w_1 = w_2 \in (1, 2)$	PT-hop discounted
3	$w_1 = w_2 = 2$	Direct $=$ PT-hop
4	$w_1 = w_2 > 2$	PT-hop prioritized
5	$w_2 > w_1 > 1$	All tiers active

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Evaluation Setup

Data & Instance

- Done on Washington D.C.
- Demand: LODES OD commute flows
- Candidate stations + Trip logs: Capital Bikeshare
- PT network: WMATA GTFS

Evaluation Questions

- **Behavioral validity:** Do w_1, w_2 change the solution in expected directions?
- **Sensitivity:** Do solutions remain stable under noisy OD demand?

Key idea

Optimize on **general mobility demand**, evaluate on **real bike trips**.

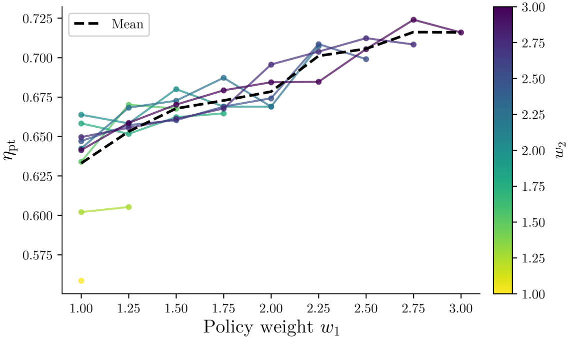
Next Steps

- Model distance decay
- Model station balance
- We found new data from Gdansk $\rightarrow$ might refine the model based on it

Thank You!

Questions?

Results: Behavioral Validity



η_{pt} across the (w_1, w_2) grid; each line is a fixed-w_2 slice.

Partial Spearman Correlations

Correlation	r_s	p
$w_1 \rightarrow \eta_{cov}$	-0.10	0.50
$w_1 \rightarrow \eta_{pt}$	$+0.88$	< 0.001
$w_2 \rightarrow \eta_{cov}$	$+0.18$	0.23
$w_2 \rightarrow \eta_{pt}$	$+0.14$	0.37

Results: Sensitivity Analysis